TASK 3: PYTHON LIBRARIES

Python Programming Practical Report — 20 Marks

1. Aim

To demonstrate the use of Python's random, math, and datetime libraries by generating a random number, performing a mathematical operation, and displaying the current date and time.

2. Problem Statement

Write a Python program that imports the random, math, and datetime libraries, uses each library for its intended purpose, and displays the results with clear output messages.

3. Libraries and Concepts Used

Library	Purpose	Function / Example
random	Generates a random number	random.randint(1, 100)
math	Performs mathematical calculations	math.sqrt(25)
datetime	Gets the current date and time	datetime.datetime.now()

4. Source Code

```
# Task 3: Python Libraries

# random is used to generate random values
import random

# math is used for mathematical operations
import math

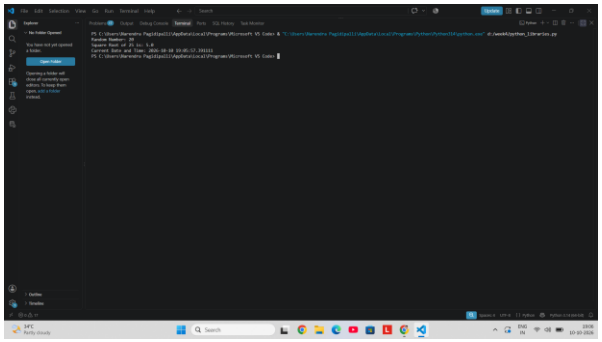
# datetime is used to work with dates and times
import datetime

# Generate a random integer from 1 to 100
num = random.randint(1, 100)
print("Random Number:", num)

# Calculate the square root of 25
number = 25
result = math.sqrt(number)
print("Square Root of", number, "is:", result)

# Display the current date and time
current = datetime.datetime.now()
print("Current Date and Time:", current)
```

5. Output



6. Project Explanation

1. The random library provides functions for generating random values. randint(1, 100) returns an integer from 1 through 100.
2. The math library provides mathematical functions. sqrt(25) calculates the square root of 25 and returns 5.0.
3. The datetime library provides date and time classes. datetime.datetime.now() returns the computer's current local date and time.
4. Comments beginning with # explain the purpose of each library and the operations performed.

7. Challenges Faced

A possible challenge is remembering which library provides each function. Importing each library and adding comments makes the program easier to read and understand.

8. Conclusion / Result

The program successfully demonstrates the random, math, and datetime libraries by generating a random number, calculating a square root, and displaying the current date and time.